

Table Cape VR Education Technology Project: An Immersive Learning Experience of Walking with Giants

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ABSTRACT

This paper profiles the Table Cape VR Education Technology Project titled "Walking with Giants," which was designed to offer an immersive learning experience for community storytelling. Table Cape is located in Wynyard, a town in rural Northwest Tasmania, Australia; it is an extinct volcano renowned for its rich tourism and heritage values. As part of a wider education technology and digital tourism initiative, Walking with Giants provides educators, students, and community members with the opportunity to visualise the high-fidelity 3D scanned terrain of Table Cape in virtual reality. Currently, the project is being exhibited at the Wonders of Wynyard Exhibition and Visitor Information Centre - a joint initiative involving creators and researchers from the University of Tasmania, industry partners from Business Northwest, and the Waratah Wynyard Council. In this paper, we describe the technical aspects of this digital technologies project and discuss the specialised knowledge-based requirements for teaching and learning incorporating such resources.

CCS CONCEPTS

• **Applied Computing**; • **Human-centered computing**;

KEYWORDS

Virtual Reality, Education Technology, User Experience, Community Engagement

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1 INTRODUCTION AND BACKGROUND: DIGITAL CREATIVITY

The Walking with Giants project forms part of an ongoing collaboration between education, training, tourism, and business interests in Northern Tasmania. The wider project seeks to respond to the affordances provided by advancing technologies, changes in the Australian Curriculum Technologies, and the changing face of tourism, as travellers seek to understand destinations in more nuanced and personal ways. From an educational perspective, the project is yielding outcomes that can be incorporated into teaching and learning resources, facilitating the training, reskilling, and upskilling of pre-service teachers and educators. The Australian Curriculum Technologies (v9) [Australian Curriculum, 2023] states that "Creating solutions for preferred futures is the overarching core concept", and it emphasises the creation of digital solutions utilising emerging technologies. From a community and tourism perspective, our tourism colleagues describe how people have become more technologically astute, with a desire to understand and learn about destinations more closely aligned with technology. Travellers increasingly want more than a map, a brochure, or a sign to understand the local culture and people. They need to feel engaged, and connected to the people, stories and culture of a destination. The opportunity to combine novel, multi-sensory technology-based tourism experiences with local stories, information and insight, allows for delivery of tourism experiences that are engaging and stimulating across a broad audience. For the community, the opportunity to showcase stories, information and passions to travellers using technology, supports a connection to visitors in

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more meaningful ways, and to experience their own stories and locale through a tourism technology lens, which can renew a sense of pride and wonder at the place they live. Technology based tourism assets therefore can provide both education and wonder for visitors and locals alike.

One of the questions raised at the start of this education technology (Edu Tech) initiative, was “how and in what ways might this VR-playable experience deepen our knowledge and understanding of the place and topic”, with relevance to STEM (Science Technologies Engineering and Mathematics) and HASS (Humanities and Social Sciences) education. Thus the project began with two main goals – first we wished to explore the utilisation of extended reality (XR) technologies for improved community engagement for placemaking, and secondly we wanted to see how our new knowledge, acquired from research into teaching and learning of technologies education, might be translated and applied to various professional learning and pre-service-teachers (PST) settings, aligned with the approaches of TPCK/TPACK [Mishra and Koehler, 2006] and SAMR [Puentedura, 2010]. One of the first practical outcomes of this project, “Walking with Giants”, is now being exhibited at the Wonders of Wynyard Exhibition and Visitor Information Centre - it focuses on creating an immersive virtual reality (VR) simulation for learning and placemaking, aiming to provide an inclusive and accessible environment for community users to engage with digital content. The Walking with Giants concept also provides school teachers and students with new possibilities for excursion activities, and creative ways to communicate education storytelling elements using VR as a medium.

In this paper, aligned with two of our main goals of the work, we will:

- Describe the technologies aspects for putting together the Table Cape Educational VR project, and

- Discuss the teaching and learning requirements regarding what and how educators can manage and teach these processes and production skills, encompassing specialised knowledge and skill sets.

2 TECHNOLOGIES ASPECTS

For the development of the Table Cape VR prototype, we introduced a high-fidelity 3D scanned miniaturised Table Cape terrain as an integral part of the educational user experience (UX). This achievement was the result of a co-creation process, in which the project team experimented with emerging technologies and digital tools to assemble the high-fidelity and high-precision 3D scan of the terrain in VR, as shown in Figure 1. As part of the UX design considerations, users (learners) will have the ability to pick up a magnifying glass and explore various points of interest and miniaturised multimedia elements that we have included in the interactive learning journey. Learners will also have the freedom to walk around and visualise the extinct volcano (Table Cape), prompting them to raise new questions about the landscape and places within the scanned terrain. The VR prototype is designed to be operated on a Real-time Ray-tracing (RTX) enabled SteamVR-based configuration, preferably with a wireless setting. This allows us to visualise the miniaturized Table Cape terrain in a highly realistic simulation, enhancing the overall user experience [Ochs and

Sonderegger, 2022] with immersive multisensory and visual realism. From a learning and teaching perspective, the technological innovations that have enabled high-fidelity representations of place and space are likely to provide distinct benefits for learners both in terms of engagement and potential educational outcomes. Firstly, the links between visual fidelity, immersion, presence, intrinsic motivation, and cognitive engagement are well established both in the broader and educational specific literature [Pausch et al., 1997][Barab et al., 2007][Neulight et al., 2007]. In essence, the more closely virtual environments approximate reality, the deeper the engagement is with the educational content. Secondly, the introduction of object manipulation and subsequent interaction between 3D objects and the simulated environment aligns closely with the principles of active learning, linked to enhanced understanding and retention of information and concepts [Balleck, 2006]. Finally, greater visual fidelity enables more authentic portrayals of place and space which have an important function both in terms of user engagement and the capacity to accurately represent geological formations and historical context.

As demonstrated in Figure 2, we tested with an experimental workflow, and innovated a hybrid approach of fusing a high-resolution map, provided by the Waratah Wynyard Council, with the scan. This allowed us to combine the repaired 3D scanned mesh (3D shape) with a precision-based map for perfecting the accuracy of the scan. The initial scan was done using over one hundred drone imageries, managed using Reality Capture and Blender, to perfect the texture of the map of the scanned terrain. This allows learners to visualise even the smallest details recorded on the map. For instance, the buildings depicted in Figure 2 are from a tulip farm, which holds significant educational and tourism value for Table Cape.

To improve the interactive learning experience of this VR project, we have also experimented with an AI-assisted character creation workflow, using Reallusion engine [Thaler et al., 2020] [Reallusion, 2023]. Figure 3 shows how Robert Jackson, an historical character who was the first Superintendent Head Lightkeeper at Table Cape Lighthouse (around 1888), was introduced in the VR environment. An audio element is also included to communicate Jackson’s story when learners are exploring the VR experience. Learners get to examine the reconstructed miniaturised light house as well, with the heritage structure reproduced digitally using some of the preserved historical data. We further introduced First Nations elements [Tasmanian Government, 2023] in the virtual space, allowing learners to find out more about the daily life of the first peoples of Table Cape, the Tommegine.

3 TEACHING AND LEARNING REQUIREMENTS

In developing Walking with Giants, the education project team from the University of Tasmania and their academic collaborators considered some of the key aspects that could be introduced in the teaching and learning of Digital Technologies units/courses within initial teacher education (ITE) programs, with a primary focused on upskilling and reskilling the PSTs. The Australian Curriculum Technologies [Australian Curriculum, 2023] suggests that

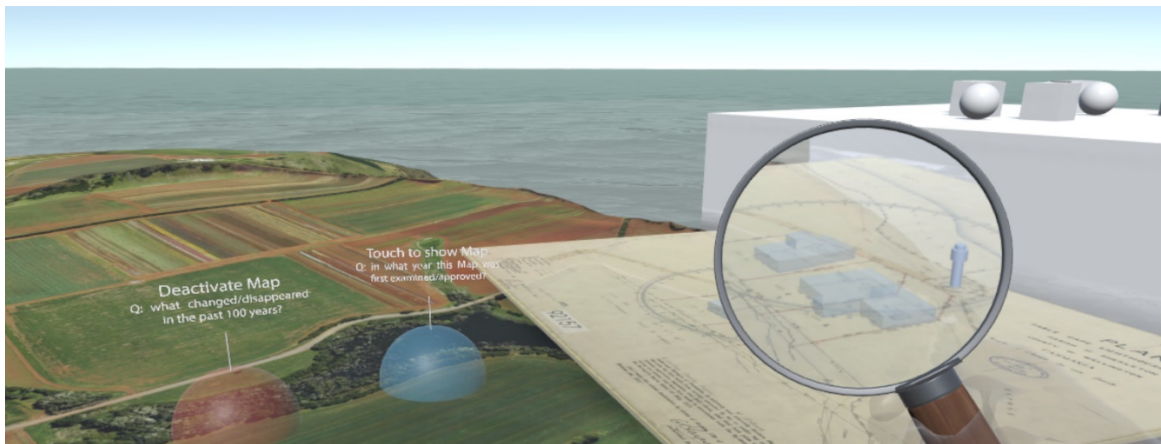


Figure 1: Table Cape VR prototype

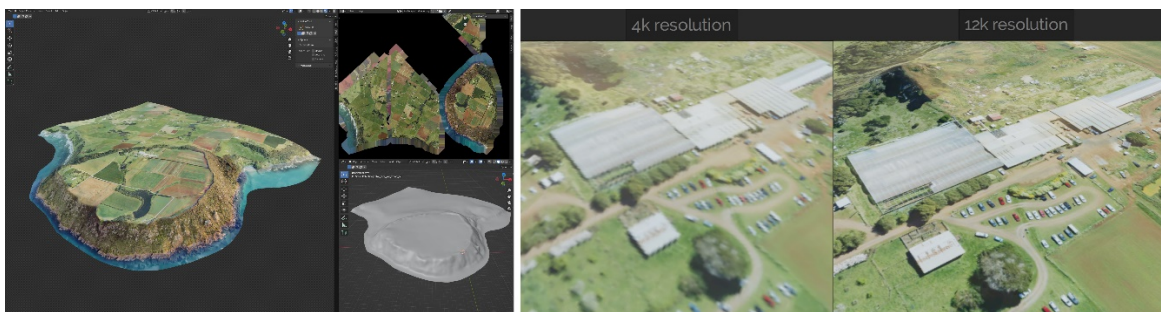


Figure 2: Hybrid approach of fusing a high precision map on the 3D scanned mesh.

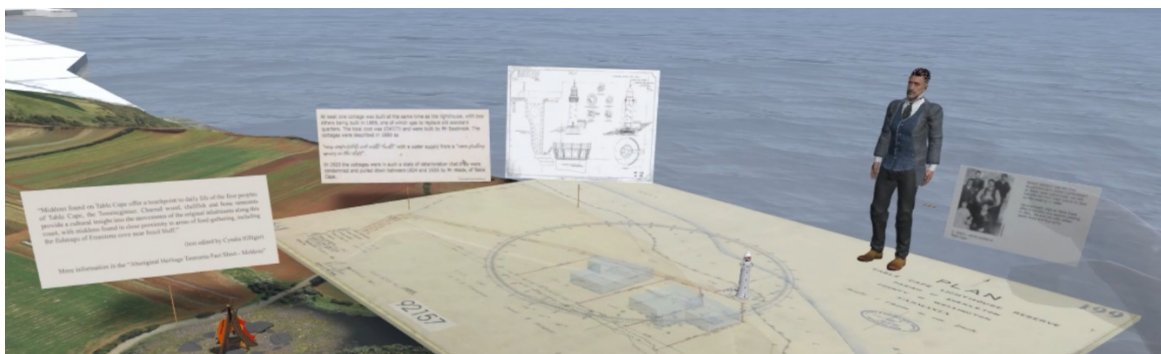


Figure 3: Hybrid approach of fusing a high precision map on the 3D scanned mesh

F-10 (foundation to year 10) / K-12 (kindergarten to year 12) teachers will: work with students, including but not limited to selecting and using emerging digital tools and advanced features to create and communicate interactive content for a diverse audience (learning descriptor AC9TDI10P11); and evaluate existing and student solutions against the design criteria, user stories, possible future impact and opportunities for enterprise (learning descriptor AC9TDI10P10); and further, that teachers and students will also

select and use advanced or unfamiliar features of digital tools to efficiently complete tasks, as part of the Digital Literacy development policy. Such descriptions may apply equally to educators outside Australia.

One study [Bower et al., 2020] suggested that the use of immersive virtual reality in education has the potential to enhance immersion, improve spatial capabilities, promote empathy, increase motivation, and possibly improve learning outcomes. However, creating such content appears to be challenging for any individual

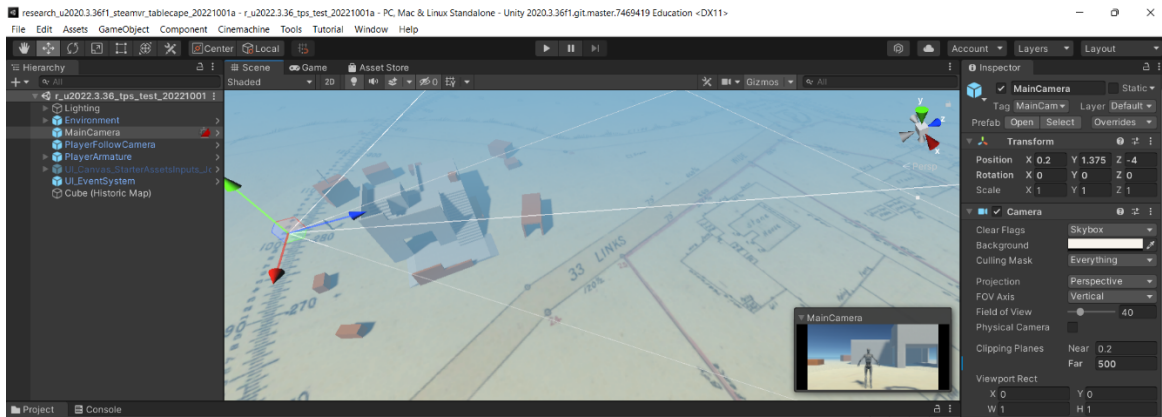


Figure 4: Level Design and Virtual Environment (VE) Recreation using Game Engine

with no pre-assumed knowledge and skillsets. For school students (guided by the teachers) to create a digital solution similar to the Table Cape Educational VR project, we propose several key requirements, for classroom teaching and learning focused on Digital Technologies working with XR and simulation learning:

Level Design and Virtual Environment (VE) Recreation using Game Engine (for Serious Games): Level design is a series of production processes in putting together a virtual environment [See et al., 2023], which allows teachers and students to work collaboratively in a classroom environment. Figure 4 demonstrates a playable experience example of a digital space reconstructed, aided with a heritage map for learning history and local places. Such an experience permits learners to control a game character and walk around the virtual world freely and ask new questions about the learning aspect of the landscape or the simulated environment. The example shown in Figure 4 was accomplished using Unity Engine, which is easily accessible by teachers and students in a classroom scenario. We suggest learning how to work with level design would be one of the easiest ways to start with for creating a simulated virtual environment. Using level design, science students could visualise earth and space simulations using serious games, or maths students could visualise employ problem-solving using a playable simulation.

3D Scanning and 3D Modelling Basics: 3D scanning technologies are now more accessible, with the introduction of LiDAR enabled devices such as Apple iPhones and iPads, or mobile applications such as Polycam and Kiri Engine usable on commonly available mobile devices. Conventionally, photogrammetry 3D scanning is only achievable using a complexed workflow of stitching together multiple photographs taken using a high-quality camera instrument. However, LiDAR enabled devices have allowed 3D scanning of educational objects/subjects, such as Figure 5 that shows the 3D scanned heritage artefact of ‘Travellyn Cairn’, which was scanned and digitised using a hand-held device within a short period of time, with the educational 3D model uploaded to the public domain of Sketchfab for knowledge dissemination. 3D modelling skills would be also required for the purpose of building a simple virtual environment, and also for repairing a 3D scanned artefact. Unity Game Engine has some built-in 3D modelling functions, and Blender is one of the external tools for doing professional grade 3D modelling

for creating high precision computer aided design content. To put it into an educational context, students could for example visualise digital maps and 3D scanned terrains in Geography.

Character Creation and Multimedia Authoring: To create an educational playable experience such as the Table Cape VR project, it would be essential for teachers and students to work with various multimedia authoring processes – including but not limited to video editing, user interface design, photo editing, and more. One of the challenging aspects in level design (or game design for creating serious games content) is creating digital characters. In the recent years, several digital tools companies such as Adobe and DAZ3D have provided creators with various options to create digital characters within a short amount of time. For Table Cape Educational VR project, we have experimented with the Reallusion AI-assisted workflow, either using some of the preserved photographs data of historic individuals, or using the appearance adjustment tools in the software, to recreate historic characters digitally. Figure 6 shows a historic character reproduced for the Table Cape 1888 environment. Such digital characters could potentially be made playable in an educational game play simulated environment, controllable by a learner/user similar to the setting shown in figure 4. Animating and controlling the digital characters, however, is one of the high-level aims, and it can be a steep learning curve for educators to manage this in a teaching and learning environment. For such a specialised knowledge, it would be best to work collaboratively with other experts with the right skills to leverage resources in an area of growth, focused on XR and simulation learning technologies.

4 DISCUSSION AND ONGOING INITIATIVE

The Walking with Giants project was made possible through the collaboration of colleagues from the University of Tasmania, Business Northwest, and the Waratah Wynyard Council. Figure 7 shows members of the public using the Table Cape VR prototype at a public launch event, enabling participants to experience educational elements significant to the local community in Tasmania.

As described in the introduction, Walking with Giants has remained accessible to the public after the launch, at the Wonders of Wynyard Exhibition and Visitor Information Centre. This has provided people from the wider community with the opportunity to

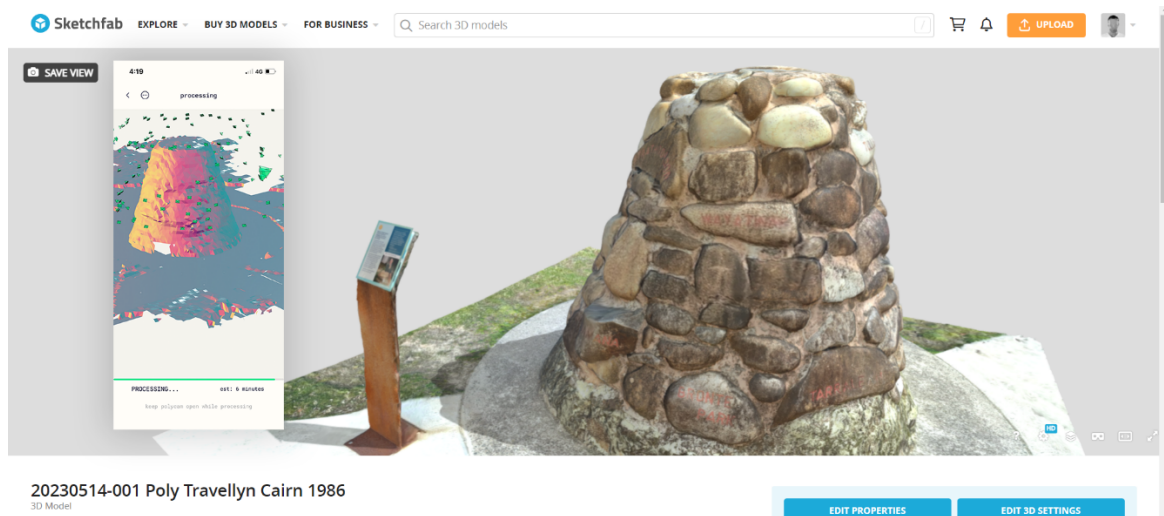


Figure 5: 3D scan example of a local heritage artefact in a public space (Tasmania), produced using Polycam.

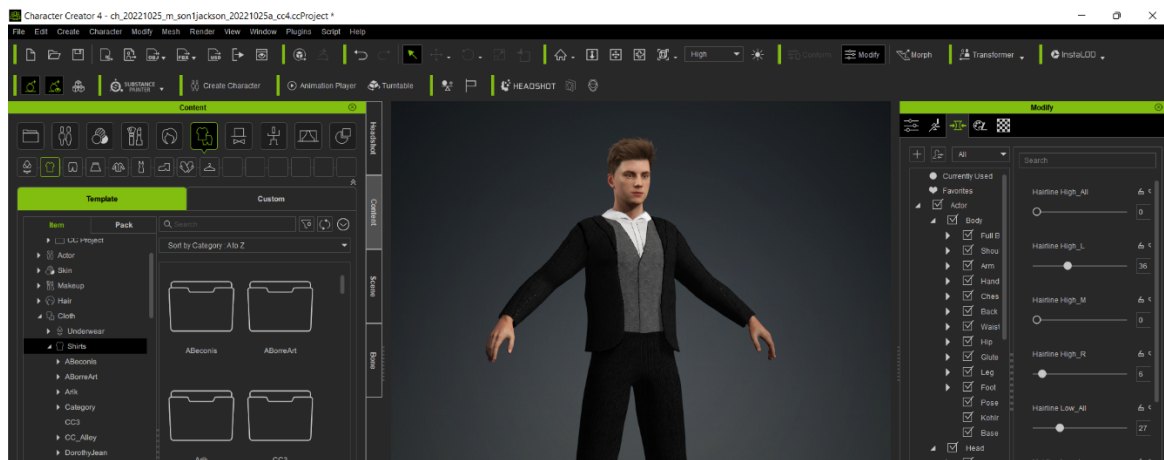


Figure 6: AI-assisted character creation workflow using Reallusion Engine.



Figure 7: Educational community engagement event at Wonders of Wynyard Exhibition and Visitor Information Centre (30 June 2023)

experience technology-enhanced content, enabling them to interact with each other, discuss new topics, and raise new questions as they engage with the educational VR experience. A touching example of the potential community impact comes from a local community caregiver, who took one of her clients to experience the Table Cape VR assets at Wonders of Wynyard: This client, very rarely speaks at all, preferring to focus on his colouring in books. After experiencing the VR technology, he spent the whole walk back to the care home talking about the experience and is still talking about it - I have never seen him so energised and enthusiastic about anything.

The tourism VR experience, a placemaking and community activation initiative, has clearly had a profound effect on this local person, and as a result of this feedback, we are now in the process of arranging to put the experiences in the care home for a day, to allow all the residents and their relatives to experience the Table Cape experiences together.

What we have learned from the Walking with Giants initiative will support ongoing research in the field of high-fidelity learning simulations using virtual reality (VR). The goal is to create a modular and customisable VR experience that can be applied to educational and placemaking initiatives. By providing high-fidelity and customisable VR spaces and content, such initiatives can enable creators, users, educators, and students to work with high-quality audio-visual content and experiences. This includes the ability to visualise 3D scanned educational artifacts with great accuracy and detail, as well as access multimedia content that is visually clearer and optically sharper within the virtual environment. Examples of educational objects and elements that benefit from high fidelity include scientific elements (e.g., geological data) and digital heritage artifacts (e.g., historic objects requiring precise archival records). This research-creation project aims to generate interest and provide benefits to educational researchers and developers who focus on utilising computer graphics and human-computer interaction (HCI) within learning technologies. In a wider perspective, the integration of technology-based subjects into various curriculum settings [Department for Education UK, 2022][Ministry of Education Singapore, 2022][National Council for Curriculum and Assessment Ireland, 2022] necessitates educators' continuous upskilling and reskilling. This growth-oriented approach emphasises educator's ability to work confidently with advanced and emerging digital tools, fostering competence and proficiency in leveraging these tools effectively, and to support the development of digital literacy. Technology-enabled applications can seamlessly integrate into a wide range of community-based, and teaching and learning activities, empowering tourism operators, educators and students to cultivate digital literacy skills and engage with emerging technologies for immersive learning experiences and the creation of XR digital content. By harnessing technology, STEM and HASS educators and students gain the ability to visualise scientific concepts through XR and learning simulations, while also delving into humanities subjects through technology-enhanced content. For instance, VR simulations of local settlements can provide historical and geographical context, immersing learners in a place-based simulated learning environment. For creators and educators, it is also essential to consider the relationship between placemaking, community activation, and the impact-driven design and computational thinking processes - these storytelling and educational narratives

should drive the responsible use of technology. As part of this ongoing initiative, the project team will continue to explore various innovative ways to work with emerging technologies, and to find out what and how some of these technologies can be adaptable in some of the teaching and learning scenarios, both for teaching how these XR content can be developed, and for using XR technologies in various educational, training, tourism and business contexts.

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REFERENCES

- Australian Curriculum. 2023. Retrieved from <https://v9.australiancurriculum.edu.au>
- Punya Mishra and Matthew J. Koehler. 2006. Technological Pedagogical Content Knowledge: A Framework for Teacher Knowledge. *Teachers College Record*, Vol. 108(6). 1017–1054. <https://doi.org/10.1111/j.1467-9620.2006.00684.x>
- Carli Ochs and Andreas Sonderegger. 2022. The Interplay Between Presence and Learning. *Frontiers in Virtual Reality*, Vol. 3. <https://doi.org/10.3389/frvir.2022.742509>.
- Markus Thaler, Stephan Schlögl and Aleksander Groth. 2020. Agent vs. Avatar: Comparing Embodied Conversational Agents Concerning Characteristics of the Uncanny Valley, 2020 IEEE International Conference on Human-Machine Systems (ICHMS), Rome, Italy. <https://doi.org/10.1109/ICHMS49158.2020.9209539>
- Reallusion. 2023. Headshot - AI-powered 3D Head Generator. Retrieved from <https://www.reallusion.com/character-creator/headshot>.
- Tasmanian Government. 2023. Aboriginal Middens (AHT Fact Sheet - Middens). Retrieved from <https://www.aboriginalheritage.tas.gov.au/Documents/AHT%20Fact%20Sheet%20-%20Middens.pdf>
- Zi Siang See, Susan Ledger, Lizbeth I. Goodman, Benjamin Matthews, Donovan Jones, Shanna Fealy, Wooi Har Ooi, Manisha Amin. 2023. Playable Experiences Through Technologies: Opportunities and Challenges for Teaching Simulation Learning and Extended Reality Solution Creation. *Journal of Information Technology Education: Innovations in Practice*, Vol. 22. 067-090.
- Matt Bower, Dorothy DeWitt, and Jennifer W. M. Lai. 2020. Reasons associated with preservice teachers' intention to use immersive virtual reality in education. *British Journal of Educational Technology*, Vol. 51. 2215-2233. <https://doi.org/10.1111/bjet.13009>
- Department for Education UK. 2022. National Curriculum in England: Design and Technology Pro-grammes of Study. Retrieved from <https://www.gov.uk/national-curriculum/key-stage-3-and-4>
- Ministry of Education Singapore. 2022. Educational Technology Journey. Retrieved from <https://www.moe.gov.sg/education-in-sg/educational-technology-journey>
- National Council for Curriculum and Assessment Ireland. 2022. Technology. Retrieved from <https://www.curriculumonline.ie/Senior-Cycle/Senior-Cycle-Subjects/Technology/>
- Ruben Puentedura. 2010. SAMR and TPACK: Intro to advanced practice. Retrieved from http://hippasus.com/resources/sweden2010/SAMR_TPACK_IntroToAdvancedPractice.pdf
- Randy Pausch, Dennis Proffitt, George Williams. 1997. Quantifying immersion in virtual reality. In *Proceedings of the 24th annual conference on Computer graphics and interactive techniques*. 13-18.
- Sasha Barab, Steve Zuiker, Scott Warren, Dan Hickey, Adam Ingram-Goble, Eun-Ju Kwon, Inna Kouper, Susan C. Herring. 2007. Situationally Embodied Curriculum: Relating Formalisms and Contexts, *Science Education* Vol. 91(5). 750782.
- Nina Neulight, Yasmin B. Kafai, Linda Kao, Brian Foley and Cathleen Galas. 2007. Children's Participation in a Virtual Epidemic in the Science Classroom: Making Connections to Natural Infectious Diseases, *Journal of Science Education and Technology* Vol. 16(1), 4758.
- Barry Balleck. 2006. Teaching for the future: Experiential learning and the use of new strategies and techniques available to the International Studies Classroom. *Conference Papers, International Studies Association Annual Meeting*, 1-16.